

BG770x-GL&BG95xA-GL

RF FTM Application Note

LPWA Module Series

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About the Document

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1 Introduction

This document describes the AT commands that are used to test the transmitting and receiving performances of Quectel BG77xA-GL and BG95xA-GL modules in RF FTM (Factory Test Mode).

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL

2 Description of AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. Description of RF FTM AT Commands

2.3.1. AT+QRFTESTMODE Enable/Disable RF FTM

This command enables or disables the RF Factory Test Mode.

AT+QRFTESTMODE Enable/Disable RF FTM

Test Command AT+QRFTESTMODE=?	Response +QRFTESTMODE: (list of supported <mode>s) OK
Read Command AT+QRFTESTMODE?	Response +QRFTESTMODE: <mode> OK
Write Command AT+QRFTESTMODE=<mode>	Response OK If there is any error: ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configurations will not be saved.

Parameter

<mode>	Integer type. Enable/Disable RF FTM. <u>0</u> Disable RF FTM
--------	---

	1	Enable RF FTM
<err>	Integer type. Error code. See Chapter 4 for more details.	

NOTE

The command will not be executed if the modem is at full functionality level (**AT+CFUN=1**) or if it is in airplane mode (**AT+CFUN=4**). The modem must be switched to minimum functionality level with **AT+CFUN=0**. After reboot, the module will return to full functionality level (**AT+CFUN=1**). See **AT+CFUN** in *document [1]*.

2.3.2. AT+QRFTEST Enable RF TX Testing in FTM

This command enables RF TX testing in FTM. In addition, it is suitable for testing the truth table of the GRFC interface (see GRFC interfaces in *documents [2]* and *[3]*)

AT+QRFTEST Transmit in FTM

Test Command Returns the list of parameters supported by the Write Command whose output wave format is CW AT+QRFTEST=?	Response +QRFTEST: <mode>,<EARFCN>,<time>,<wave>,<TX_power>,<offset> OK
Write Command Output wave format: CW AT+QRFTEST=<mode>,<EARFCN>,<time>,<wave>,<TX_power>,<offset>	Response OK If there is any error: ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Write Command Output wave format: GMSK in GSM AT+QRFTEST=<mode>,<EARFCN>,<time>,<wave>,<TX_power>,<bw>	Response OK If there is any error: ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Write Command Output wave format: SC-FDMA of Cat M AT+QRFTEST=<mode>,<EARFCN>,<time>,<wave>,<TX_power>,<bw>,<MSC>,<RB_num>,<RB_pos>	Response OK If there is any error: ERROR

	If there is any error related to ME functionality: +CME ERROR: <err>
Write Command Output wave format: SC-FDMA of NB-IoT AT+QRFTEST=<mode>,<EARFCN>,<time>,<wave>,<TX_power>,<MSC>,<sub_spa>,<sub_index>	Response OK If there is any error: ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configurations is not saved.

Parameter

<mode>	Integer type. 3 LTE TX test in FTM
<EARFCN>	Integer type. EARFCN of the supported uplink channels in GSM/LTE.
	Range of Uplink EARFCN GSM band
	128–251 GSM850
	1–124, 975–1023 EGSM900
	512–885 DCS1800
	1536–1834 PCS1900
	Range of Uplink EARFCN LTE band
	18000–18599 LTE Band 1
	18600–19199 LTE Band 2
	19200–19949 LTE Band 3
	19950–20399 LTE Band 4
	20400–20649 LTE Band 5
	21450–21799 LTE Band 8
	23010–23179 LTE Band 12
	23180–23279 LTE Band 13
	23850–23999 LTE Band 18
	24000–24149 LTE Band 19
	24150–24449 LTE Band 20
	26040–26689 LTE Band 25
	26690–27039 LTE Band 26
	27040–27209 LTE Band 27
	27210–27659 LTE Band 28
	131972–132671 LTE Band 66

<time>	Integer type. TX test time. Unit: millisecond. 0 Continuous mode (always in RF TX FTM) 1 600000
<wave>	Integer type. Type of transmitted signal. 0 SC-FDMA of Cat M / GMSK in GSM 1 CW (continuous waveform) 2 SC-FDMA of NB-IoT
<TX_power>	Integer type. TX power, which is absolute output power. Unit: dBm × 100.
<offset>	Integer type. offset to central frequency. Unit: Hz.
<bw>	Integer type. Uplink tone bandwidth. It can only be set to 0. 0 1.4 MHz
<MCS>	Integer type. Modulation and Coding Scheme. Wave: SC-FDMA of Cat M Range: 0–15. Wave: SC-FDMA of NB-IoT Range: 0–12.
<RB_num>	Integer type. Number of RB allocation. Range: 1–6.
<RB_pos>	Integer type. The position of RB allocation. Range: 0–5. The value of <RB_pos> is related to the value of <RB_num>, and <RB_num> + <RB_pos> ≤ 6.
<sub_spa>	Integer type. Subcarrier spacing. 0 15 kHz 1 3.75 kHz
<sub_index>	Integer type. Subcarrier index. If <sub_spa>=0 (15 kHz), the range of <sub_index> is 0–18. If <sub_spa>=1 (3.75 kHz), the range of <sub_index> is 0–47.
<err>	Integer type. Error code. See Chapter 4 for more details.

NOTE

1. **GSM band** is valid only on BG955A-GL module.
2. The command will be valid only if **AT+QRFTESTMODE=1** is enabled.
3. For PCS1900, **<EARFCN>** should be added 1024 to distinguish PCS1900 EARFCN from DCS1800 EARFCN. PCS1900 EARFCN will be from 1536 to 1834. For example, PCS1900 EARFCN 512 is encoded as 1536 since 512 + 1024 = 1536.

2.3.3. AT+QRXFTM Enable RF RX Testing in FTM

This command enables RF RX testing in FTM, suitable for measuring the receiving signal properties.

AT+QRXFTM Receive in FTM

Test Command AT+QRXFTM=?	Response +QRXFTM: <mode>,<EARFCN> OK
-----------------------------	--

Read Command AT+QRXFTM?	Response OK
Write Command AT+QRXFTM=<mode>,<EARFCN>	Response +QRXFTM: <min_value>,<avg_value>,<max_value> OK If there is any error: ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<mode>	Integer type. 2 LTE RX test in FTM																																		
<EARFCN>	Integer type. EARFCN of the supported downlink channels in LTE. <table> <tr> <th>Range of Downlink EARFCN</th><th>LTE band</th></tr> <tr><td>0–599</td><td>LTE Band 1</td></tr> <tr><td>600–1199</td><td>LTE Band 2</td></tr> <tr><td>1200–1949</td><td>LTE Band 3</td></tr> <tr><td>1950–2399</td><td>LTE Band 4</td></tr> <tr><td>2400–2649</td><td>LTE Band 5</td></tr> <tr><td>3450–3799</td><td>LTE Band 8</td></tr> <tr><td>5010–5179</td><td>LTE Band 12</td></tr> <tr><td>5180–5279</td><td>LTE Band 13</td></tr> <tr><td>5850–5999</td><td>LTE Band 18</td></tr> <tr><td>6000–6149</td><td>LTE Band 19</td></tr> <tr><td>6150–6449</td><td>LTE Band 20</td></tr> <tr><td>8040–8689</td><td>LTE Band 25</td></tr> <tr><td>8690–9039</td><td>LTE Band 26</td></tr> <tr><td>9040–9209</td><td>LTE Band 27</td></tr> <tr><td>9210–9659</td><td>LTE Band 28</td></tr> <tr><td>66436–67335</td><td>LTE Band 66</td></tr> </table>	Range of Downlink EARFCN	LTE band	0–599	LTE Band 1	600–1199	LTE Band 2	1200–1949	LTE Band 3	1950–2399	LTE Band 4	2400–2649	LTE Band 5	3450–3799	LTE Band 8	5010–5179	LTE Band 12	5180–5279	LTE Band 13	5850–5999	LTE Band 18	6000–6149	LTE Band 19	6150–6449	LTE Band 20	8040–8689	LTE Band 25	8690–9039	LTE Band 26	9040–9209	LTE Band 27	9210–9659	LTE Band 28	66436–67335	LTE Band 66
Range of Downlink EARFCN	LTE band																																		
0–599	LTE Band 1																																		
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9040–9209	LTE Band 27																																		
9210–9659	LTE Band 28																																		
66436–67335	LTE Band 66																																		
<min_value>	Integer type. Minimum value of the result of the received power range. Default Value: 0. Unit: dBm.																																		
<avg_value>	Integer type. Average value of the results of the received power range. Default Value: 0. Unit: dBm.																																		
<max_value>	Integer type. Maximum value of the result of the received power range.																																		

Default Value: -90. Unit: dBm.

<err>

Integer type. Error code. See **Chapter 4** for more details.

NOTE

1. This command will be valid only if the **AT+QRFTESTMODE=1** is enabled.
2. The result of the Write Command will represent the current values of the received signal properties.
3. When applying CW to UE antenna, it is recommended to use 1 MHz offset to central frequency to avoid DC interference.
4. When applying LTE signal to UE antenna, it is recommended to use a continuous FDD radio frame, which occupies all subcarriers including the ones dedicated for PBCH/PSS/SSS.

3 Examples

3.1. Enable RF FTM

```

AT+CFUN=0 //Enable minimum functionality level
OK

AT+QRFTESTMODE=1 //Enable RF FTM
OK

AT+QRFTESTMODE? //Query the current RF FTM state of the module
+QRFTESTMODE: 1

OK

```

3.2. Test TX in RF FTM

```

//Output wave format: CW
AT+QRFTEST=3,18300,0,1,2300,0 //RF TX uplink channel 18300 for LTE Band 1, TX power
                                //is 23 dBm
OK

AT+QRFTEST=3,21625,0,1,2300,0 //RF TX uplink channel 21625 for LTE Band 8, TX power
                                //is 23 dBm
OK

//Output wave format: GMSK in GSM
AT+QRFTEST=3,512,60000,0,3000,0 //RF TX uplink channel 512 for DCS1800, TX power
                                //is 30 dBm
OK

//Output wave format: SC-FDMA of Cat M
AT+QRFTEST=3,18300,0,0,2300,0,0,6,0 //RF TX uplink channel 18300 for LTE Band 1, TX power
                                //is 23 dBm
OK

AT+QRFTEST=3,21625,0,0,2300,0,0,6,0 //RF TX uplink channel 21625 for LTE Band 8, TX power
                                //is 23 dBm

```

OK

//Output wave format: SC-FDMA of NB-IoT

AT+QRFTEST=3,18300,0,2,2300,0,0,0 //RF TX uplink channel 18300 for LTE Band 1, TX power is 23 dBm

OK

AT+QRFTEST=3,21625,0,2,2300,0,0,0 //RF TX uplink channel 21625 for LTE Band 8, TX power is 23 dBm

OK

3.3. Test RX in RF FTM

AT+QRFTESTMODE=1 //Enter RF FTM

OK

AT+QRXFTM=2,300 //Enable RF RX downlink channel 300 for LTE Band 1
+QRXFTM: 0,-52,0

OK

AT+QRXFTM=2,900 //Enable RF RX downlink channel 900 for LTE Band 2
+QRXFTM: 0,-90,0

3.4. Disable RF FTM

AT+QRFTESTMODE=0 //Disable RF FTM

OK

//After any type of RF test a reboot is required.

4 Summary of CME ERROR Codes

The CME ERROR code **<err>** indicates an error related to mobile equipment or network. The details about **<err>** are described in the following table.

Table 3: Related CME ERROR Codes

<err>	Meaning
501	Full Functionality Mode enabled
502	RF FTM is not enabled

5 Appendix References

Table 4: Related Documents

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual
[2] Quectel_BG770A-GL&BG773A-GL_Hardware_Design
[3] Quectel_BG95xA-GL_Hardware_Design_V1.0

Table 5: Terms and Abbreviations

Abbreviation	Description
CW	Continuous Waveform
DC	Direct Current
EARFCN	E-UTRA Absolute Radio Frequency Channel Number
FDD	Frequency Division Duplexing
FTM	Factory Test Mode
LPWA	Low-Power Wide-Area
LTE	Long Term Evolution
MCS	Modulation and Coding Scheme
NB-IoT	Narrow Band Internet of Things
PBCH	Physical broadcast channel
PSS	Primary Synchronization Signal
RB	Resource Block

RF	Radio Frequency
RX	Receive
SC-FDMA	Single-carrier Frequency-Division Multiple Access
SSS	Secondary Synchronization Signal
TX	Transmit
UE	User Equipment
